

Name: _____

Date: _____

TRIGONOMETRY, (SINE AND COSINE RULES)

LO: I can apply the sine rule and cosine rule to find unknown sides and angles in non-right-angled triangles.

Sine and cosine rules

The **sine rule** is used when you have a side and its opposite angle: $a/\sin A = b/\sin B = c/\sin C$. The **cosine rule** is used when you know two sides and the included angle (or all three sides).

Cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$. To find an angle: $\cos A = (b^2 + c^2 - a^2) \div 2bc$.

Quick examples

●	Sine rule: $a/\sin A = b/\sin B$	for matching pairs of side and angle
●	Cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$	SAS or SSS
●	Use sine rule for AAS or ASA	when you can match a pair
●	Sine rule works with two sides and included angle	use cosine rule for SAS

Key idea

Sine rule for matched pairs (side + opposite angle). Cosine rule for SAS or SSS.

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

Each side is proportional to the sine of its opposite angle

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – sine rule finding a side

In triangle ABC, $A = 40^\circ$, $B = 70^\circ$, $a = 8$. Find b .

$$a/\sin A = b/\sin B$$

$$8/\sin 40^\circ = b/\sin 70^\circ$$

$$b = 8 \sin 70^\circ / \sin 40^\circ = 11.7 \text{ (1 dp)}$$

$$\boxed{b = 11.7}$$

Cross-multiply to find the unknown side.

Example 2 – sine rule finding an angle

In triangle, $a = 10$, $b = 7$, $A = 50^\circ$. Find B .

$$\sin B / b = \sin A / a$$

$$\sin B = 7 \sin 50^\circ / 10 = 0.536$$

$$B = \sin^{-1}(0.536) = 32.4^\circ$$

$$\boxed{B = 32.4^\circ}$$

Use the sine rule flipped to find the angle.

Example 3 – cosine rule finding a side

$b = 5$, $c = 7$, $A = 60^\circ$. Find a .

$$a^2 = 5^2 + 7^2 - 2(5)(7)\cos 60^\circ$$

$$= 25 + 49 - 70(0.5) = 39$$

$$a = \sqrt{39} = 6.2 \text{ (1 dp)}$$

$$\boxed{a = 6.2}$$

Substitute into the cosine rule formula.

Example 4 – cosine rule finding an angle

$a = 8$, $b = 6$, $c = 5$. Find angle A .

$$\cos A = (b^2 + c^2 - a^2) / 2bc$$

$$= (36 + 25 - 64) / (2 \times 6 \times 5) = -3/60$$

$$A = \cos^{-1}(-0.05) = 92.9^\circ$$

$$\boxed{A = 92.9^\circ}$$

A negative value of $\cos$ means the angle is obtuse.

YOUR TURN – PRACTICE (deliberate practice)

1. Sine rule - finding sides

Use the sine rule to find the unknown side (1 dp).

- a) $A = 50^\circ$, $B = 60^\circ$, $a = 10$. Find b .
 b) $A = 30^\circ$, $C = 80^\circ$, $a = 6$. Find c .
 c) $B = 45^\circ$, $C = 75^\circ$, $b = 9$. Find c .
 d) $A = 40^\circ$, $B = 55^\circ$, $b = 12$. Find a .

2. Sine rule - finding angles

Use the sine rule to find the unknown angle (1 dp).

- a) $a = 8$, $b = 6$, $A = 50^\circ$. Find B .
 b) $a = 12$, $b = 9$, $A = 70^\circ$. Find B .
 c) $b = 10$, $c = 7$, $B = 65^\circ$. Find C .
 d) $a = 15$, $c = 11$, $A = 80^\circ$. Find C .

3. Cosine rule - finding sides

Use the cosine rule to find the unknown side (1 dp).

- a) $b = 7$, $c = 9$, $A = 50^\circ$. Find a .
 b) $a = 6$, $c = 10$, $B = 80^\circ$. Find b .
 c) $a = 5$, $b = 8$, $C = 120^\circ$. Find c .
 d) $b = 4$, $c = 6$, $A = 45^\circ$. Find a .

4. Cosine rule - finding angles

Use the cosine rule to find the unknown angle (1 dp).

- a) $a = 7$, $b = 5$, $c = 8$. Find A .
 b) $a = 10$, $b = 8$, $c = 6$. Find A .
 c) $a = 4$, $b = 5$, $c = 6$. Find C .
 d) $a = 9$, $b = 7$, $c = 5$. Find A .

5. Mixed problems

Choose and apply the correct rule (1 dp).

- a) $A = 35^\circ$, $B = 85^\circ$, $a = 7$. Find b . b) $a = 10$, $b = 12$, $C = 70^\circ$. Find c .
 c) $a = 8$, $b = 9$, $c = 11$. Find largest angle. d) $B = 40^\circ$, $b = 8$, $c = 10$. Find C .
 e) $a = 5$, $c = 7$, $B = 100^\circ$. Find b .

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) The sine rule uses matching pairs of side and opposite angle

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Correct answer: _____

Reason: _____

b) Use the sine rule when you know two sides and the included angle

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Correct answer: _____

Reason: _____

c) $a^2 = b^2 + c^2 - 2bc \cos A$ is the cosine rule

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Correct answer: _____

Reason: _____

d) The cosine rule has a plus sign: $a^2 = b^2 + c^2 + 2bc \cos A$

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Correct answer: _____

Reason: _____

e) If $\cos A$ is negative, then angle A is obtuse

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Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Two ships leave a port. Ship A sails **12 km** on bearing **060°**. Ship B sails **8 km** on bearing **130°**. Find the distance between the two ships.

Expression: _____

Simplified: _____

b) In triangle PQR, $PQ = 9 \text{ cm}$, $QR = 11 \text{ cm}$, and angle $PQR = 75^\circ$. Find PR. Then find angle QPR.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
 Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

